



The convolutional neural networks (CNNs) contained in new automotive computers must be trained to locate and classify important objects and to determine their distance from the vehicle. Training requires presenting the CNNs with tens of thousands of images containing the objects and supplying their true classifications and distances

systems (ADAS) that rely on a range of sensors to detect and classify objects, helping to make split-second decisions. These sensor systems include various types of cameras and radar, but they face a big challenge: performing effectively when it is dark, which is a significant factor in the recent rise in nighttime incidents. We are all in this together, working on solutions to keep our streets safer, especially when the sun goes down.

In a recent chat, Nidhi Agarwal - co-author of this article, spoke to Wade Appelman, Chief Business Officer at Owl Autonomous Imaging, about how thermal imaging technology can make a big difference in nighttime pedestrian safety within the automotive industry.

How smart cameras make cars safer?

In the discussion, they concluded that it was all about the perfect blend of hardware and software that was driving some amazing breakthroughs in this field. At the heart of this innovation is a thermal camera that works hand in hand with advanced software modules known as convolutional neural networks (CNN). These components are up and running smoothly on Nvidia

Jetson Orin processors, making it a practical and efficient solution for real-world applications. Let's dive into the important components and capabilities:

Smart neural networks. The software in these cars can do some pretty amazing stuff like classifying images, segmenting them, and even determining distances. It is like having a super-smart co-pilot that understands everything around you in real time.

ROS integration. These cars talk

in a language called 'ROS topics,' making it easier to turn complex data into actionable insights. Think of it as the car translating its surroundings into helpful advice for safer driving.

Versatile outputs. Imagine your car having multiple vision modes—from basic thermal views to 3D object listing, and even combining regular and thermal imaging. It's like having night vision goggles, giving you a clear view in low light and detailed colour information.

Efficient data management.

These cars are not just smart, they're efficient too! They focus on processing only the important stuff, which means quicker decisions and less power use. It's all about being fast and efficient without draining the battery.

Monocular vision in imaging technology

Imagine having just one eye but still being able to judge distances and depths accurately. That is essentially what monocular vision technology does. It takes a single image from a camera and, using some clever

Car companies adapting to new safety rules

To all car enthusiasts and concerned citizens, we are at a pivotal moment in the automotive industry. Faced with strict government safety mandates and the pressing need to protect pedestrians, automakers are navigating a challenging path. Here is a brief look at this evolving landscape:

Cost concerns. The industry, traditionally cautious about costs, is being pushed to innovate while keeping production viable.

Safety mandates. Global governments are stepping up with regulations that focus not just on passenger safety but also on protecting pedestrians.

Historical lessons. Remember how seatbelts and airbags became standard? They were also born from regulatory pressures.

Technological leap. Automakers are turning to technologies like thermal imaging to meet these new safety standards—a potential game-changer for night-time pedestrian safety.

Future outlook. By 2028, we might see a significant shift with many vehicles equipped with this life-saving technology.

As stakeholders in road safety, whether as drivers, pedestrians, or enthusiasts, we are witnessing a crucial transition. Automakers, balancing costs and compliance, are embracing innovation to make our roads safer for everyone. Let's stay tuned to see how this journey unfolds, driving us toward a safer future.